Advantages

1. It has the ability to store more energy than steel and iron springs.
2. It will not break suddenly like metallic springs.
3. The life of the spring is higher.
4. Rubber can withstand vibrations very well.

7.1.3 Air Spring

The layout of air spring is shown in figure 7.1.3(1&2). Air spring gives more comfort and smooth ride to the travellers. It provides more comfortable ride when the vehicle is unloaded and loaded conditions. They are classified into two types, namely,

7.1.3 1. Bellow Type
2. Piston Type

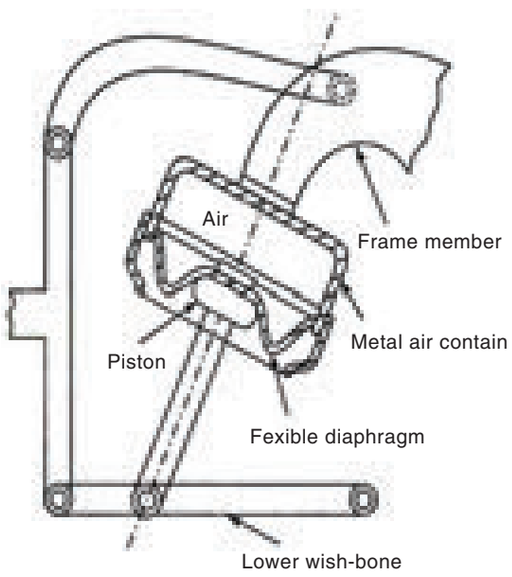


Figure 7.1.3 (2) Piston Type

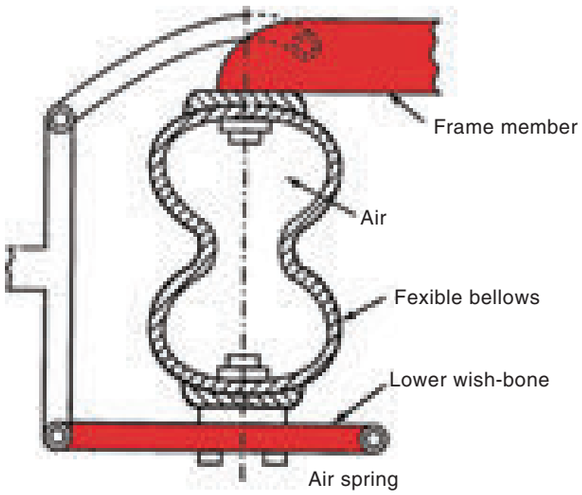


Figure 7.1.3 (1) Bellow Type

7.2 Helper Spring

This system is widely used in heavy commercial vehicles, trucks. This helper spring does not function during light load condition. As the load increases, beyond a certain level, the spring will start functioning. The helper spring is integrated along with the Master leaf in leaf spring. Due to the increased load, the master leaf spring and helper leaf gets fully expanded and the helper spring touches the special bracket, makes the helper leaf to function. This type of helper leaf is widely used in the rear axle.

7.3 Types of suspension system

1. Conventional suspension system
2. Front or Rear independent suspension system
3. Air suspension system
4. Hydro elastic suspension system

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